03-7 The Mean Formula

1. Arithmetic Mean

$$\bar{X} = \mu = \frac{\sum_{i=1}^{n} X_i}{n}$$

2. Geometric Mean

$$\bar{X}_g = \mu_g = \left(\prod_{i=1}^{n} X_i\right)^{1/n}$$

3. Harmonic Mean

$$\bar{X}_{Hr} = \frac{n}{\sum_{i=1}^{n} 1/X_i}$$

→ F_1 is harmonic mean of Precision and recall

$$F_1 = \frac{2}{\frac{1}{prec} + \frac{1}{rec}} = \frac{2\,prec \times recall}{prec + recall}$$

Harmonic Mean

$$\mu_{Hr} = \frac{n}{\sum_{i=1}^{n} \frac{1}{X_i}} = \frac{n \prod_{i=1}^{n} X_i}{\sum_{j=1}^{n} \prod_{i \neq j} X_i}$$

4. Root Mean Square

$$RMS = \sqrt{\frac{\sum X_i^2}{n}}$$

6. Relationship between harmonic mean

$H \leq G \leq A \leq R$ $\quad G^2 = A \cdot H$

$\mu_H \leq \mu_g \leq \mu_A \leq RMS$ $\quad \mu_g^2 = \mu_A \cdot \mu_H$